Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Python · LangGraph · Full-Stack Engineering · AWS · Automation Pipelines

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | McLean area, Virginia

PROFESSIONAL SUMMARY

AI engineer and software developer with 8+ years building production-grade automation systems and ML pipelines in regulated financial environments. Built a **LangGraph-powered AI agent pipeline** (LeaseI) processing 1,240 compliance documents — **93.6% precision, 87.9% F1**, every determination traceable to its triggering rule. At Entrata, encoded manual compliance review logic into automated pipelines eliminating **80% of eligibility workload** with zero violations, and shipped full-stack property management solutions (PHP, JavaScript, SQL, REST APIs, Docker) at **99.5% uptime across 7+ years**. Designed ML pipelines achieving **$R^2 = 0.977$** across 190+ countries (Python, R) and Isolation Forest anomaly detection surfacing outliers manual review missed. Strong PropTech domain — 7 years building compliance and analytics infrastructure for the same property portfolios Valon services. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 – Aug 2024

Analytic Software Engineer — Full-Stack Engineering · Automation Pipelines · PropTech Compliance

- Built and maintained scalable full-stack property management solutions (PHP, JavaScript, SQL, REST APIs, Docker-based CI/CD) across **100K+ unit portfolios** — **99.5% uptime over 7+ years** of production operation; shipped automation tooling used daily across Section 8/HCV, LIHTC, and HUD compliance workflows.
- Encoded manual income eligibility review logic into automated compliance pipelines — AMI threshold calculations, recertification workflows, escalation routing across **millions of tenant certifications**; eliminated **80% of manual review workload** with zero compliance violations.
- Migrated all client data to **AWS S3** with Libsodium encryption — **100% PHI/HIPAA compliance**, zero security incidents; optimized **200+ SQL queries for 30% performance improvement** across 1M+ property records.
- Built **property performance dashboards** tracking eligibility rates, waitlist demand, occupancy, rent burden vs. AMI, delinquency, and lease renewal trends across 100K+ units; conducted 300+ peer code reviews and mentored 5+ junior engineers.

George Mason University

Aug 2024 – Aug 2025

Research Associate — ML Pipelines · AI-Powered Anomaly Detection · Data Analytics

- Built end-to-end ML pipelines (Python, R) on WHO + UK Road Safety datasets (190+ countries) — regression models achieving **$R^2 = 0.977$** with feature engineering, cross-validation, and automated reporting.
- Applied **Isolation Forest AI anomaly detection** across 3 research pipelines — surfaced outliers manual review missed; deployed self-service dashboards (Tableau, ggplot2, Quarto) improving faculty research comprehension **40%**.

PROJECTS

LeaseI: AI Agent Compliance Pipeline

Python · LangGraph · RAG · Scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an end-to-end **LangGraph AI agent workflow** processing 1,240 compliance documents — document extraction, NLP classification, and deterministic audit engine achieving **93.6% precision, 87.9% F1**; every compliance determination traceable to its triggering rule with full metadata lineage.

CrashClarity: Large-Scale ML Classification (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Mined 104K+ UK accident records (Python, Scikit-learn) — built Random Forest + Logistic Regression ensemble with feature engineering across weather, road, and temporal variables; achieved **85%+ classification accuracy**; deployed interactive dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · ML pipelines, anomaly detection, statistical modeling, data governance, AI systems

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

ML & AI: Python, R, LangGraph, RAG, NLP, LLMs, AI Pipelines, Scikit-learn, TensorFlow, XGBoost, Random Forest, Logistic Regression, Isolation Forest, Anomaly Detection, Feature Engineering, Pandas, NumPy

Engineering & Cloud: PHP, JavaScript, TypeScript, FastAPI, REST APIs, Docker, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), Linux, Git

Data & Databases: SQL, PostgreSQL, MySQL, PL/SQL, Oracle, SQLite, ETL Pipelines, Data Governance, PHI/HIPAA Compliance, Metadata & Lineage

BI & Visualization: Tableau, Power BI, ggplot2, Quarto, matplotlib, Jupyter, Interactive Dashboards

Process: Agile/Scrum (120+ sprints), Audit Pipelines, PropTech/Compliance Domain (LIHTC, HCV, HUD), Zero-Incident Data Migrations